Engineering Portfolio

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Product Design | R&D | Additive Manufacturing | Computational Modeling

Who am I?

I am a self taught, independent engineer, and founder of Sanchez Performance LLC. I am obsessed with designing and building systems. I finished high school at an accelerated pace and forwent the traditional college because that allowed me to dedicate my time fully to engineering. I focused on real-world design, prototyping, and testing. Learning through direct application rather than theory alone. Failure has become the most satisfying part of my day because it means there is something for me to improve somewhere. I have *actual experience*, designing, building, breaking, iterating, and optimizing products for reliability, strength, and cost.

Philosophy

Engineering is a closed loop process: design a thing, build it, break it, analyze it, then repeat (at least) a dozen times. Failure and its mechanisms, force us to confront real world constraints, creating data points to iterate on. Failure then, by nature, is multidisciplinary. That's what it asks of you, to think on the level of interacting systems; as an engineer your job isn't just to design, it's to evaluate feasibility, manufacturability, cost, repeatability, and understand the geometry of those obstacles. Failure is data. Data is king.

Skill and Toolset

Design & Simulation

- CAD (Solid modeling, assemblies, tolerancing)
- Engine simulation software
- Flow modeling
- Theoretical thermodynamics
- BMEP & VE modeling

Manufacturing

- FDM additive manufacturing
- Carbon fiber reinforced polymers
- Annealing & dimensional compensation
- Geometry Optimization for Manufacturability
- Post-processing & coatings

Project List

Pneumatically Actuated Atomization for Evaporative Cooling in an Automotive Context

Water/Methanol Injection is a mechanism by which liquids with favorably high specific heat capacities and subsequently evaporative cooling properties are introduced into the intake system of an engine to lower charge air temperatures. This is usually adapted to engine systems which utilize forced induction, in which intake air is compressed above atmospheric pressure(by various means), to improve volumetric efficiency above 100%. Due to the ideal gas law, as intake air is compressed, it also rises in temperature; water/methanol injection is used primarily in race/motorsport settings to counteract this and improve efficiency by maintaining lower air temperature and subsequently air density.

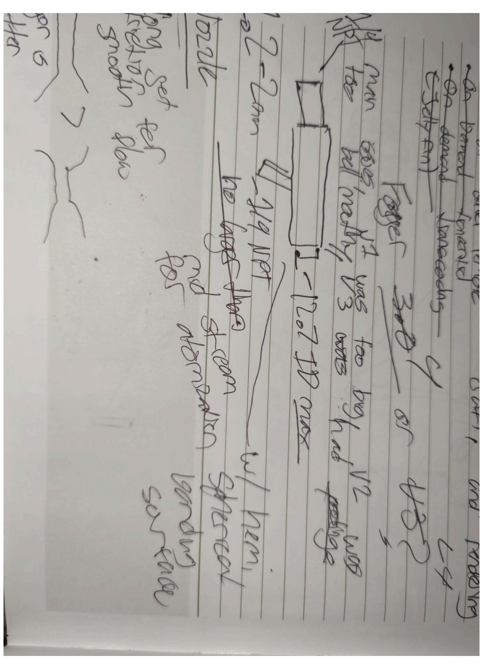
Problems:

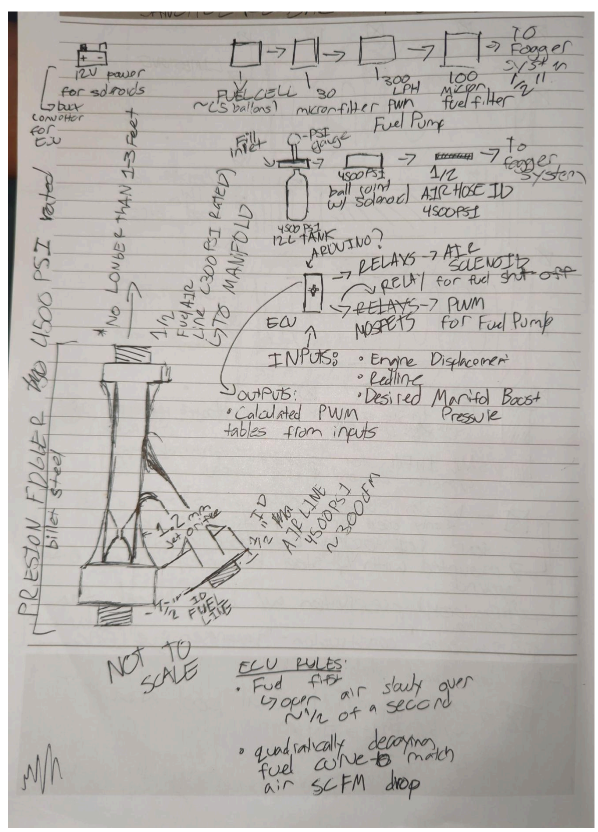
- **Pump dependency**: Traditional water/methanol systems rely on electrically driven pumps, increasing complexity, cost, and failure points.
- **Chemical variability**: Injection systems often encounter ethanol, gasoline, or isopropyl alcohol. Materials must resist corrosion and chemical degradation.
- **Complex wiring**: PWM-controlled pumps require intricate wiring and control logic, raising implementation difficulty and cost.

Objectives:

- Reduce cost and complexity
- Remove PWM pump dependency
- Achieve equal or greater atomization for heat transfer
- Make implementation simple and universal

First Drawings/Brainstorming





First Iteration:

- Initially only pneumatically **assisted for atomization**
- Still required pressurized liquid chamber
- Qualitative Testing, only comparing atomization to current market nozzles.

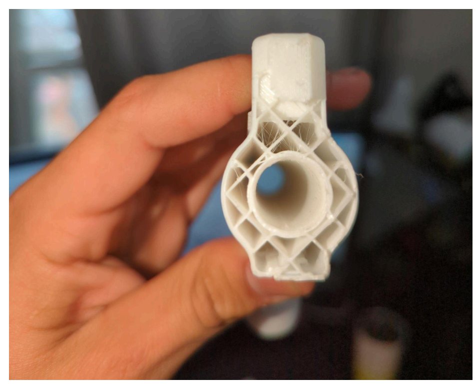
Mainly a proof of concept for additively manufactured injection nozzles.

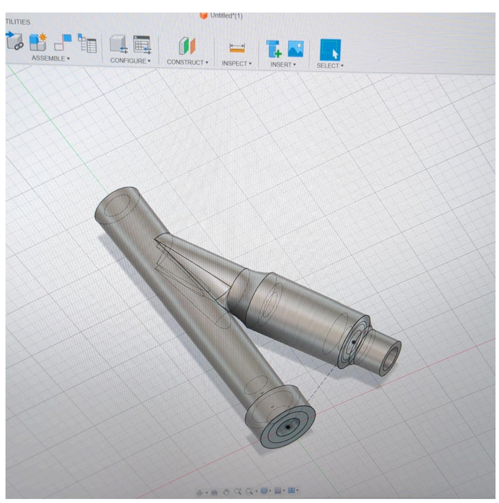
Problems:

- The prototype was made out of PLA, and would eventually degrade with sufficient chemical exposure and engine bay heat.
- Print orientation caused threads to shear in testing; air hose whipped abruptly when disconnected and sheared threaded areas off.
- Micro pores in between layers retained water in the part itself and would leak out.

Successes:

- Atomization was excellent
- Additive manufacturing proved to be ideal for altering nozzle and internal flow paths with ease and accuracy.





Analysis:

- Failure was along layer lines
- Internal infill pattern reveals 'stringyness', suggesting inadequate filament humidity which worsened print quality and strength.
- Future Prototypes need more wall layers, denser and/or altered infill patterns. To improve strength and seal printed areas to prevent absorption of liquid and leaking.

Second Generation(Multiple Iterations)

- Switched to Nylon-612 filaments for chemical resistance
- Used chopped carbon fibers to improve printability and dimensional stability
- Now fully-venturi based. Internal nozzle geometry and flow path was tweaked to create pressure differentials and move liquid solely with vacuum motive. (No longer required pressurized liquid chamber)

Ver 2.1

Ver 2.2

Ver 2.3

This generation was entirely focused on experimentation with nozzle geometry to optimize vacuum strength.

2.1 and 2.2 were the only versions to have sufficient vacuum in the liquid channel to promote atomization. Each iteration in this generation tweaked the distance of insertion of the liquid channel from the air inlet. This was done to evaluate where the lowest inlet air pressure section was. 2.1 and 2.3 both utilized straight, uniform air channels and did not manipulate air pressure through choked orifices. 2.2 had a slightly tapered diffuser, however the angle of the taper was too shallow to be effective or cause meaningful change.

Problems:

- Subpar print quality from moisture absorption, inadequate print settings, and outside geometry not optimized for best surface finish.
- 2 of 3 iterations cracked during initial testing, when installing 1/4NPT hose couplings. This was because the nozzle inserts weren't tapered like the threads on the hose coupling, so the nozzle throat expanded and cracked the body of the nozzle when fully inserted.
- Atomization was fine enough, too many large droplets*, sporadic, not reliable enough.

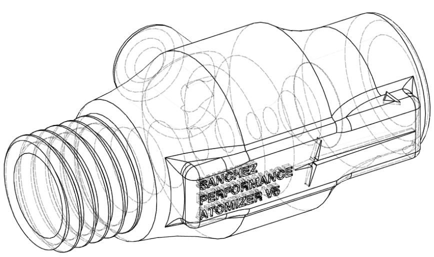
2.2 used an epoxy filler to seal the crack for temporary testing.

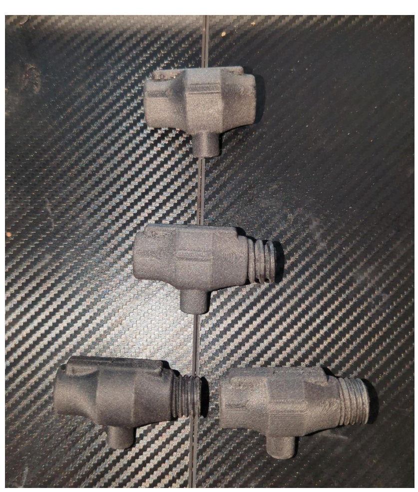
*Large droplets are not ideal because they can cause liquids to pool in the intake and in turn overcool the engine, leading to issues like knock.

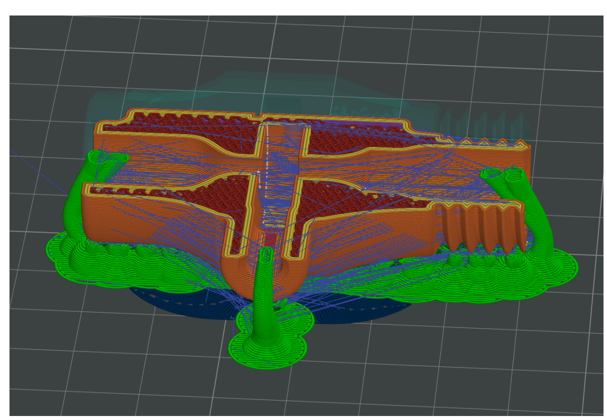
Third Generation(Multiple Iterations)

- Resolved cracking issues by tapering inlets to match ¼ NPT thread specifications
- Improved print quality and settings with proper filament preparation and active moisture management systems
- Incorporated more walls(3-8 depending on version) and switch to gyroid infill for isotropic strength improvements
- Used choked flow analysis and increased liquid inlet diameter.

(versions are 3.1-3.4 from top-down and left-right)







The final internal geometry for the nozzle was refined and established in this generation. Most iterations focused on determining which type of threads would be strongest and improving print orientation so that most shear stress moments were as perpendicular as possible to printed layer lines. Version 3.2 finalized taper geometry for ¼ NPT threads. All units were pressure tested from 5 to 200 PSI and did not experience any damage or failures due to pressure. This layer orientation proved to be strong for threaded sections.

Pressure Dynamics Analysis overview:
Choked Flow Calculation(Rough Estimates):

- Upstream: 1/4" NPT, 200 psi compressed air
- Orifice: 3-5 mm diameter (depending on version)
- Discharge coefficient: 0.85
- Resulting choked mass flow rate: 0.054 kg/s
  Produces fine atomization of water/methanol into the expansion chamber.

Expansion Chamber Vacuum Estimate:

- Chamber volume designed to allow rapid expansion and droplet breakup
- Estimated pressure drop relative to ambient: 1–3 kPa
- Promotes evaporation and prevents droplet coalescence

Estimated Air Speed at Orifice: ~324 m/s (choked, near sonic)

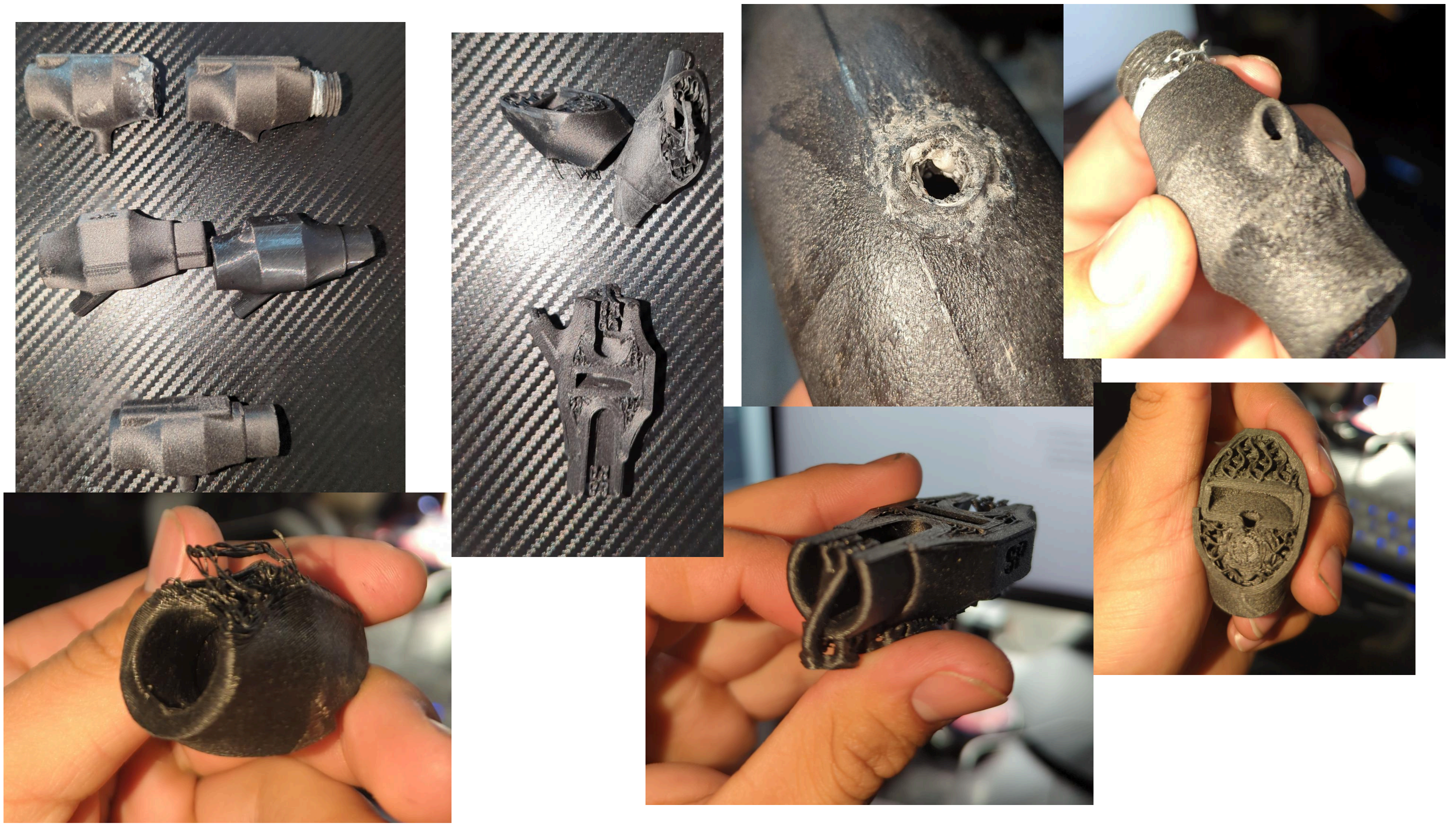
- Produces strong atomization into the expansion chamber
- High momentum aids in dispersing water/methanol droplets before evaporation

Problems

- Vacuum ejector geometry meant that past 130PSI and 3-4 CFM, backpressure was greater than what the diffuser throat could accommodate for; vacuum pressure above 100PSI dropped, and at 130PSI the venturi effect of the ejector was no longer operating and the nozzle did not function. This quickly became an issue without regulated inlet air.
- Print times were too long due to excessive infills and walls on some versions.
- None of these nozzles have been heat soak tested so heat resistance is still unknown.

3.5th Generation (Multiple iterations)

- First generation to be engine bay tested and fully heatsoaked for a 4 week long period.
- Adopted 1/4 liquid inlets as opposed to 1/2 to maintain vacuum across longer hose run



This generation of iterations had the most failures.

1. Mid-printing failures occurred because of layer adhesion issues with Polythalamide, (low chamber temperature meant poor inter-layer crystal diffusion). PPA was adopted in an attempt to combat fragility challenges and polymer creep in the ¼" versions, however it was proving to be weaker and difficult to work with. Polymer creep became an issue in nozzles subject to engine bay heat soak, causing them to deform when clamp style installation was first attempted(with Nylon 612).

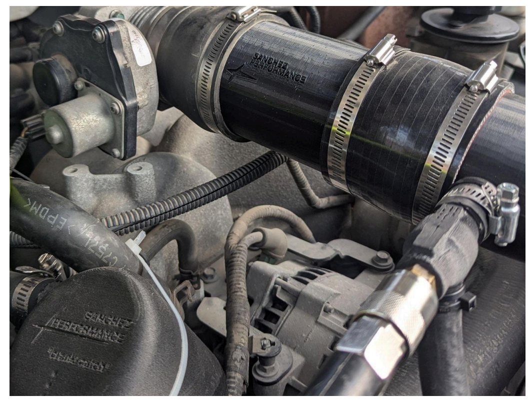
2.¼" Liquid inlets snapping off when they were still made of Nylon-612. Changed liquid channel angle and print orientation to be perpendicular to any twisting forces.

However, these angle changes altered the pressure dynamics of the nozzle and the vacuum was no longer strong enough to promote any atomization.

3. When the original straight liquid channel version print settings were altered to increase interlayer cross sectional area on the liquid inlet and prevent snapping, the thread print orientation was offset at the same angle, so when first installed onto an intake, the threads themselves snapped off of the nozzle body.

Fourth Generation (Final)

- Resolved layer adhesion issues with Polythalamide-CF, so now ¼ and ½” nozzles were adequately strong.
- Switch to securing clamps in favor of threads for reliability and universal installation.
- Resolved polymer creep by making PPA-CF print more reliably and adopting a preheating technique to ensure chamber temperature is optimal.
- Reverted back to print orientation from the Third Generation.
- Nozzles impregnated with nanoceramics(polysilazane) for sealing and surface finish improvements
- Used mechanical air regulators for simplicity.



Overview on Objective Achievement

- PWM pump dependency removed using clever vacuum ejector nozzle geometry.
- Total cost to manufacture is $1 dollar in filament + $2-3 in NPT couplings + $0.10 on nanoceramic coating. Compared to $100-130 retail for current market options: https://kombustionmotorsports.com/products/aem-v3-water-methanol-injection-nozzle-only-kit-qty-2
- Atomization is superior due to near-mach choked flow characteristics
- Wiring is simplified, just a DC solenoid to control airflow.
- Installation is universal, clamps onto a ⅝” bung on intake tubing.

Surface Area and Flow Optimized Condensing Surfaces for Engine Crank Case Pressure Relief Applications

High-performance and forced induction engines often experience blow-by, where combustion gases escape past the piston rings and enter the crankcase. These gases carry oil mist, fuel vapors, and other contaminants into the intake system if not properly managed.

Problems caused by unmanaged blow-by:

- Intake contamination: Oil deposits on throttle bodies, intake runners, and intercoolers reduce efficiency and heat transfer.
- Fuel dilution: Oil mist in the intake can lower octane locally and contribute to knock.
- Engine wear: Recirculated oil vapors increase carbon buildup and accelerate wear in valves and pistons.

Traditional catch cans are often passive, relying on coalescence of oil droplets via baffles or mesh filters. They are limited by flow capacity, prone to clogging, and may require frequent maintenance.

Objective: Develop an additively manufactured catch can capable of-

- Efficiently separating oil and fuel vapors under high CFM intake conditions
- Handling pulsating crankcase blow-by without backpressure issues
- Being additively manufactured with tunable internal geometries
- Minimizing maintenance despite being additively manufactured.

First Generation: Proof-of-Concept

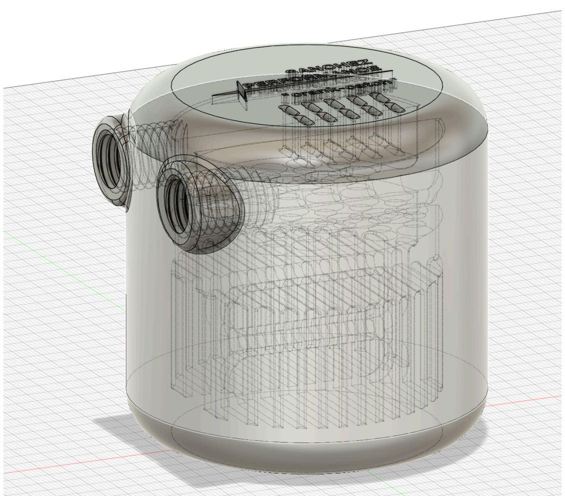
- Material: PLA prototype for initial feasibility
- Internal design: 2 stage baffle, one to slow down incoming blow-by gasses and the third being a printed mesh layer to improve fuel, and oil vapor condensation in the catch can.
- Goals: Test flow, droplet separation, and manufacturability of complex internal baffles.

Problems:

- PLA prototype is easier to print than high chemical resistance polymers that would be used in later generations, so manufacturability is still somewhat uncertain
- Baffle geometry ineffective at high CFM, allowing oil carryover, because the mesh was only a single
- Threads and ports were not fully evaluated for strength yet.

Successes:

- Verified that additive manufacturing enables rapid iteration of internal geometries
- Confirmed feasibility of integrating multi-port, multi-baffle design into a compact housing



Second Generation: Geometry Optimization

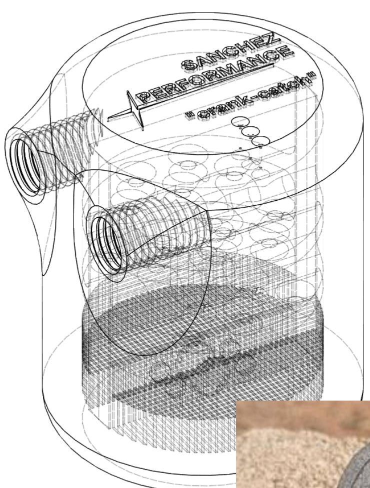
- Material: Nylon-612-CF for chemical resistance
- Internal design: three-stage baffles with perpendicular vanes to promote oil coalescence
- Added modular ports for intake and crankcase hoses
- Implemented vibration-resistant thread sealant

Problems:

- One iterations cracked at port inserts under torque from hose fittings
- Oil separation improved, but high-flow pulses could still cause some bypass, Mainly due to perpendicular vanes being too restrictive. Still OK for low blow-by situations.

Successes:

- Demonstrated that baffle angle and spacing significantly influence oil capture efficiency
- Confirmed ability to tune internal channels to prevent oil carryover even with harder to print filaments like Nylon 612-CF

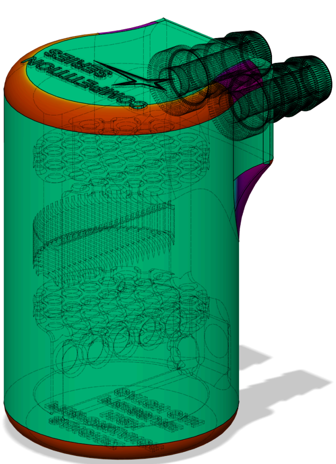



Fourth Generation: Engine-Ready & Heat-Soaked

- Material: PPA-CF for heat and chemical resistance
  Nanoceramic (polysilazane) coating to prevent vapor absorption and improve surface finish
- Print-in-Place hosebarbs replaced threads and screw-in fittings, making it a unibody design, improving strength.
- Internal channels fine-tuned for high CFM blow-by pulses and minimized droplet re-entrainment, even with increased baffling layers.
- Added 10mm drain plug for improved serviceability and cleaning.

Achievements:

- Engine bay tested for 4+ weeks under continuous heat soak
- Modular, universal fitment for ⅜” to ¾” hoses
- High separation efficiency at high blow-by rates without backpressure issues




Iterative Hyper-Accelerated Aerodynamic Solver

High-fidelity aerodynamic simulations are critical in performance automotive design, but **traditional CFD is slow and computationally expensive**, often taking hours or days for a single geometry iteration. This limits rapid prototyping, testing, and optimization of complex components like splitters, diffusers, and wings, when precision isn't required.

Objective: Develop an aerodynamic solver that:

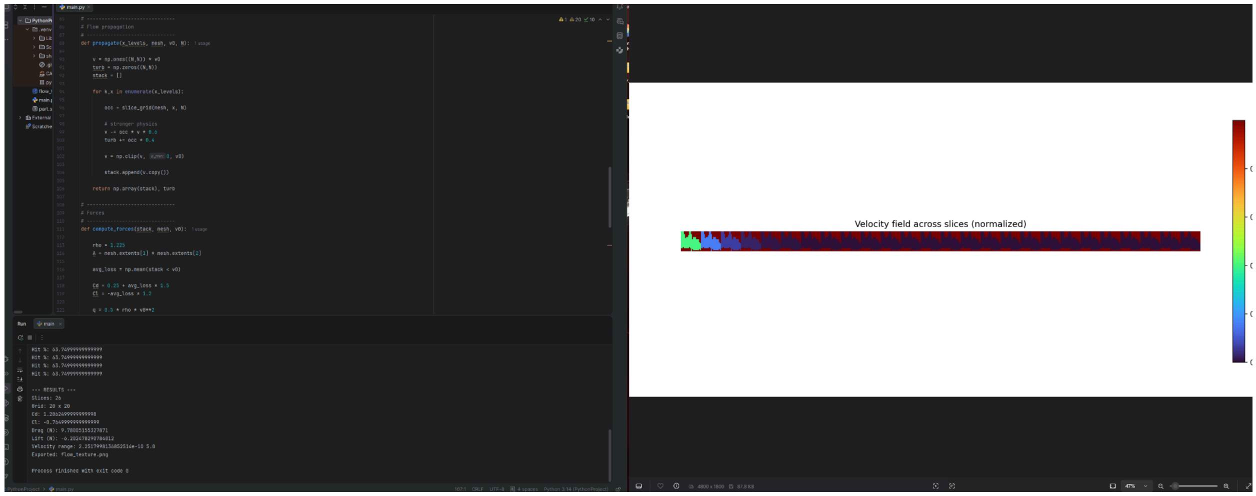
- Dramatically reduces simulation time while maintaining a lower resolution of predictive accuracy
- Enables rapid design iterations for complex 3D geometries
- Integrates with additive manufacturing workflows for quick prototype validation

Problems with Traditional CFD:

- Long computation times (hours to days per simulation)
- High hardware requirements for transient or turbulent flows
- Limited ability to test multiple design iterations in a practical timeframe for quick prototyping.

Specifics:

- **Hyper-accelerate aerodynamic simulations** using self iterative algorithms.
- **Focus computational effort** by subdividing models in relation to airflow and using genetic algorithms to maintain adequate resolution.
- Produce **predictive pressure, velocity, drag, and downforce figures** for immediate design feedback



Solution

I designed and built a custom surrogate aerodynamic solver focused on rapid iteration for performance automotive applications. Instead of relying on traditional CFD pipelines, the system uses a slice-based raycasting approach to propagate airflow through complex 3D geometries, generating real-time estimates of velocity fields, drag, and downforce at a fraction of the computational cost.

In layman's terms, it subdivides a model into, uses ray casting and predictive models that estimate the effect on the ray, iterates that through each subdivision, and uses the output data to estimate drag, downforce, and other things.

The solver automatically normalizes model scale, adapts spatial resolution based on user input, and exports low-resolution flow texture maps for visualization and analysis. This enables fast design validation within additive manufacturing workflows, allowing physical prototypes to be evaluated in minutes rather than hours or days. The result is a highly iterative, compute-efficient aerodynamic tool tailored for rapid motorsport and engineering development cycles.

Key Technologies: Python, Trimesh, numerical methods, raycasting, surrogate modeling, data visualization